

Solution Architecture — Script-Controlled ACL (Institution Details)

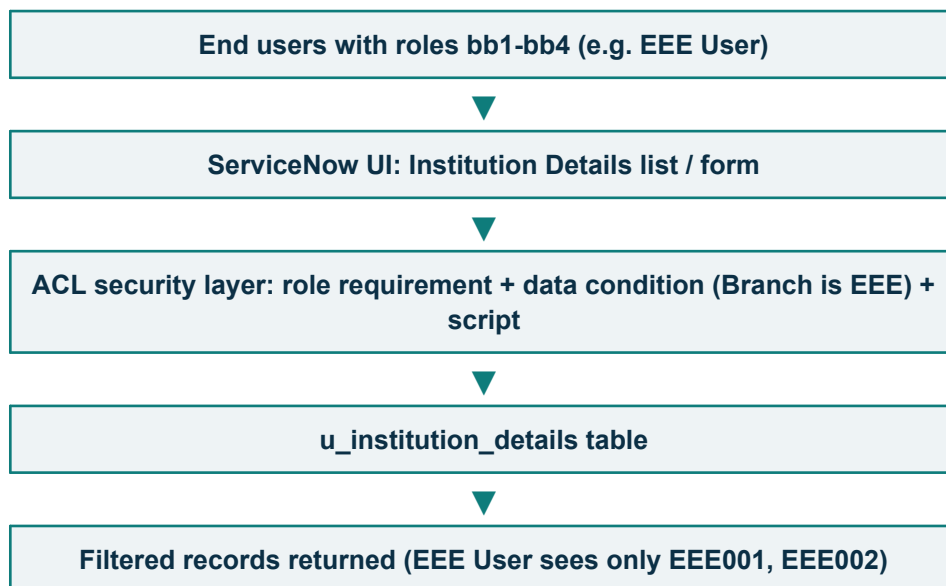
Phase 3 · Project Design Phase

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This document describes the solution architecture for the Script-Controlled ACL project on ServiceNow PDI dev443713. It shows how a request from a branch end-user flows through the ServiceNow UI into the Access Control List security layer, where a role requirement, a data condition and a server-side script together decide access to rows in the `u_institution_details` table, and how filtered records are returned. All layers are native ServiceNow platform components; there is no external application, framework or database.

Platform	ServiceNow Now Platform (Australia / Zurich family release)
Instance	dev443713.service-now.com (Personal Developer Instance)
Security engine	Access Control List (ACL) with server-side GlideScript (gs.hasRole)
Core table	u_institution_details with Branch choice field (ECE / EEE / CSE)

Architecture Flow



Component Responsibilities

Component	Responsibility
End users and roles (bb1 read, bb2 create, bb3 write, bb4 delete)	Identify the requester and the operation attempted. The test user <code>EEE User</code> holds all four <code>bb</code> roles; <code>admin</code> (System Administrator) always passes; <code>Abel Tuter</code> holds none.
ServiceNow UI / Institution Details list	Presents the list and form for <code>u_institution_details</code> and issues the read, create, write or delete request; renders only what the ACL layer permits, or the "Security constraints prevent access" message when denied.
ACL security layer (<code>sys_security_acl</code>)	For each operation, evaluates the required role, then the data condition, then the script. Read is Advanced with the condition Branch is <code>EEE</code> (<code>u_branch=EEE</code>); Create, Write and Delete check only their <code>bb</code> role.
READ ACL script (<code>gs.hasRole</code>)	Server-side GlideScript: returns true if the user has role <code>admin</code> , true if the user has role <code>bb1</code> , and false otherwise, enforcing the field-value-plus-role decision for reads.
<code>u_institution_details</code> table	Stores the Institution Details records (Student Roll Number, Student Name, Faculty Name, Branch, Email, Phone Number, Description) including <code>ECE001</code> , <code>EEE001</code> , <code>EEE002</code> and <code>CSE001</code> ; returns only rows that satisfy the ACL evaluation.
Supporting configuration (Impersonation, <code>security_admin</code> , default role removal)	<code>security_admin</code> elevation enables editing ACLs; Impersonation verifies each persona's view; removing the auto-generated <code>u_institution_details_user</code> role ensures every ACL depends only on its <code>bb</code> role.